

MDM Presentation vs API Automation Framework

Gap Analysis Report

Project: Indoore MDMS Backend API Automation

Generated: 19 June 2026

Scope: Playwright API framework only (src/modules)

MDM Presentation PDF vs API Automation Framework — Gap Analysis

Source PDF: templates/MDM Presentation_23.12.2025.pdf (131 slides, 23-Dec-2025) **Compared against:** Playwright API automation in src/modules/*/tests/*.spec.ts

Scope: Framework only — manual testing / Excel workbook **not** included.

The PDF is a screenshot deck. Text extraction yields **slide titles only** (no steps or API paths). Mapping is at **feature/screen → automation module** level.

Summary

Metric	Count
PDF named screens	~95
Automation modules	22
Spec files	127
Fully covered	~42
Partially covered	~18
Not covered	~35

Framework coverage (rough): ~63% of PDF screens have at least one API spec. Core MDM flows (auth, dashboard, master data, DTRs, consumers, billing, MIS, HES commands, energy audits) are well represented. Large report families (SAIFI/SAIDI, Collection Reports, Aberration, Billing Exchange) have **no module**.

Automation modules (framework inventory)

Module	Specs	PDF area
AUTH	11	Login, session, invite
DASHBOARD	5	Home / overview widgets
OVERALL-DASHBOARD	2	Overall overview
MASTER-DATA	4	Consumer / DTR / Feeder / Substation master
DTRS	7	DTR overview, events, daily analysis
FEEDER	4	Feeder profile, consumption, alerts
CONSUMERS	13	Consumer profile, load, events, billing
CONSUMPTION	5	Daily / pattern / net meter
BILLING	2	Day-wise and monthly billing
COMMERICAL-ANALYSIS	6	Commercial summary, PF, MD, LF, compare, pattern
TECHNICAL-ANALYSIS	2	Technical summary and analysis
MIS-DASHBOARD	16	Communication, event data, priority
ENERGY-AUDITS	7	Loss analysis, hourly loss, network trends
HES-COMMANDS	14	Meter commands, config, history, samples
REPORTS	3	Event report, event detail, DTR billing
NOTIFICATIONS	2	Web + mobile notifications
ASSET-MANAGEMENT	3	Org / network hierarchy, DTR lookup
UTILS-LOOKUP	14	Search and dropdown lookups
USERS-ADMIN	3	User management, security, devices
ROLE-PERMISSIONS	1	Role permission CRUD
MODULES-PERMISSIONS	1	Module permission CRUD
AUDIT-LOGS	2	Audit list and export

Slide-by-slide mapping (framework only)

Legend:  Covered |  Partial |  Missing

Auth & Home

PDF Slide	Framework module / spec
MDM Portal Login	✓ AUTH — login, refresh, invite flows
MDM Home Screen	⚠ DASHBOARD — metrics, DTR summary
Static Overview Dashboard	⚠ DASHBOARD + OVERALL-DASHBOARD
Overall Overview	✓ OVERALL-DASHBOARD — dashboardmetrics, dtrcommunication

Master Data & DTR

PDF Slide	Framework module / spec
MDM Consumer Master Data	✓ MASTER-DATA/consumer-master.spec.ts
DTR Overview / DTR Overview Detail	✓ DTRS — profile, statistics, feeders, events

HES / Meter Commands

PDF Slide	Framework module / spec
Meter Command — Single Meter	✓ HES-COMMANDS — meter, meter-info, query-meter-job, payment, billing
Meter Command — Bulk Meter	⚠ search-meters, history; no explicit bulk-only spec
Meter Command Configuration	✓ profile-config, demand-config, metering-mode, load-curtailement

Billing Exchange & Notifications

PDF Slide	Framework module / spec
Billing Exchange Report — valid	✗ No module
Billing Exchange Report — invalid	✗ No module
Mobile Notification	✓ NOTIFICATIONS/notificationsmobile.spec.ts

Energy Audits / Loss Analysis

PDF Slide	Framework module / spec
Network Loss Analysis — Billing	✓ ENERGY-AUDITS/loss-analysis-*.spec.ts
Consumption Compare — Hourly Loss Report	✓ ENERGY-AUDITS/hourly-loss-report-*.spec.ts
DTR IP Data / LS Data / DP Data	✓ ENERGY-AUDITS loss-analysis by data type

MIS Dashboards

PDF Slide	Framework module / spec
MIS Dashboard	✓ MIS-DASHBOARD (16 specs)
Communication Dashboard	✓ communication, communicationstats
Live Communication Dashboard	⚠ communication stats only; no live-specific spec
Communication Dashboard Monthly	⚠ No dedicated monthly spec
Event Data Phase wise	✓ eventdatavoltage, eventdatacurrent, eventdatapower
Event Data Category wise	✓ event-classification
Event Data Priority Wise	✓ eventpriority 1–6, priority-overview
Event — No Restoration	⚠ eventdatanonrollover
MIS Download	✗ No export/download API test

Commercial Analysis (22 PDF sub-screens)

PDF Slide	Framework module / spec
Commercial Analysis (summary)	✓ commercial-summary.spec.ts
PF Violation	✓ powerfactor.spec.ts
MD > CD Last Three Month	✓ mdanalysis.spec.ts (1 variant)
Sanction Load Violation	⚠ Data in mdanalysis.data.ts ; not executed in spec
LF < 5% / LF < 100%	⚠ loadfactor.spec.ts — operator: 1t , months: 1 only
LF < 5% Last 3/6 Month	⚠ months param exists; not tested
Consumption Compare Last Month	✓ consumptioncompare.spec.ts
Consumption Compare Same Month Last Year	⚠ Data exists; not in spec
Abnormal Low Consumption	⚠ Data exists; not in spec
Zero Consumption (1/3/6/9/12/>12 month)	⚠ consumptionpattern — zero , months: 1 only
100-unit kWh (3/6/9 month)	⚠ threshold param; not all month windows
LAST SIX MONTH 50% AVG < Initial	✗ No spec
Night Zero Consumption	✗ No spec
Night consumption <= 10% of Day	✗ No spec

Technical & Techno-Commercial

PDF Slide	Framework module / spec
Technical Analysis	✓ TECHNICAL-ANALYSIS (2 specs)
Techno Commercial Analysis	✗ No module

Reports — Communication & Events

PDF Slide	Framework module / spec
MIS Report	⚠️ REPORTS — eventreport, eventdetail, dtrbilling
Communication — Consumer Detail / IP / DP / LS	⚠️ CONSUMERS + HES-COMMANDS meter-samples partial
Event — Consumer Event / Detail	✓ eventreport, eventdetail
Event — DT Wise interruption Detail	✗ No dedicated spec

DTR Operations

PDF Slide	Framework module / spec
DTR Communication	✓ DASHBOARD/OVERALL dtrcommunication
DTR Event — Interruption / Detail	✓ DTRS/dtrevents
DTR Hourly Load — Actual Load / Load % / Consumption	⚠️ CONSUMERS/liveloadprofile partial
DTR Daily Analysis — Loading / Unbalance / Summary	⚠️ dtrdailythresholdchart, dtrcapacitygauge

Consumption, Billing, Pattern

PDF Slide	Framework module / spec
Consumer Consumption Hourly / Daily / Monthly	⚠️ dailyconsumption; hourly/monthly gaps
Night Zero Consumption (00:00–06:00)	✗ No spec
Consumption Monthly Net Meter	✓ monthlynetmeter
Billing — Day wise / Monthly	✓ daywisebilling, billingdata
Pattern — Last 3 Months / Yearly / Comparison	✓ lastthreemonths, yearly, comparison

Consumer Data & Collection Reports

PDF Slide	Framework module / spec
Consumer Data — Min/Max Voltage	⚠️ CONSUMERS/powerquality
Consumer Data — Current Without Voltage	✗ No spec
Collection of Report (11 variants)	✗ No module

Prepaid, Aberration, SAIFI/SAIDI

PDF Slide	Framework module / spec
Prepaid Summary	⚠ HES payment command only; no summary report API
Aberration Report / Detail / Entry / ATR Zone	✗ No module
SAIFI / SAIDI (7 report variants)	✗ No module

In framework but not prominent in PDF

Module	Purpose
USERS-ADMIN	User management, security, devices
ROLE-PERMISSIONS / MODULES-PERMISSIONS	Access control
AUDIT-LOGS	Audit list and export
ASSET-MANAGEMENT / UTILS-LOOKUP	Hierarchy and search APIs
FEEDER	Feeder profile, electrical params, daily consumption

Priority gaps (framework only)

P0 — No automation module

1. **SAIFI / SAIDI reports** (PDF slides 124–130) — 7 report types
2. **Collection of Reports** (slides 94–104) — 11 meter/alarm analysis reports
3. **Billing Exchange Report** valid/invalid (slides 11–12)
4. **Aberration Report / Entry / ATR** (slides 118–122)
5. **Techno Commercial Analysis** (slide 55)

P1 — Module exists but coverage is thin

1. **COMMERCIAL-ANALYSIS** — 22 PDF sub-screens; only ~6 specs; many data variants not executed (Sanction Load, Same Month Last Year, Abnormal Low, zero 3/6/9/12 months, LF 3/6 months, night rules)
2. **MIS Download** — no export API test
3. **Prepaid Summary** — only HES payment command, not prepaid report
4. **Live vs Monthly Communication Dashboard** — no distinct specs
5. **Consumer hourly consumption, Current Without Voltage**
6. **Event — DT Wise interruption Detail**

P2 — Quick wins inside existing modules

Expand specs using data already in `src/modules/COMMERCIAL-ANALYSIS/Data/` :

- `consumptioncompare.data.ts` — add tests for Same Month Last Year, Abnormal Low
 - `mdanalysis.data.ts` — add Sanction Load Violation, Improper MD
 - `consumptionpattern.data.ts` — add zero pattern with months 3, 6, 9, 12
 - `loadfactor.api.ts` — add months 3, 6 and operator `gt` (LF > 100%)
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Recommended next steps (framework)

1. **New modules** (need backend endpoints): SAIFI-SAIDI, COLLECTION-REPORTS, BILLING-EXCHANGE, ABERRATION, TECHNO-COMMERCIAL
 2. **Expand COMMERCIAL-ANALYSIS** — parameterized specs for all PDF commercial sub-types
 3. **Add CONSUMPTION** specs for hourly consumption and night-zero rules if APIs exist
 4. **Add REPORTS/MIS export** spec once download endpoint is confirmed
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PDF extract: `reports/mdm-pdf-extract.txt` | Automation root: `src/modules/`